

Corrective markers across languages

1. Introduction

Mandarin Chinese has two particles *bing* and *you* for correction and contradiction. In a dialogue, they may be used to correct the interlocutor, and thus require there to be a proposition to correct:

(1) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:

#Zhangsan {bing/you} bu jianshen.

Zhangsan BING/YOU NEG work.out

Intended: 'Zhangsan doesn't work out.'

Bing and *you* have to co-occur with negation and be immediately followed by it:

(2) A: 'Zhangsan works out.'

B: Zhangsan {bing/you} *(bu) jianshen.

Zhangsan BING/YOU NEG work.out

'Zhangsan doesn't work out.'

When correcting a negative proposition, *bing* and *you* still require negation, leading to a sentence with double negation:

(3) a. A: 'Zhangsan doesn't work out.'

B: *Zhangsan {bing/you} jianshen.

Zhangsan BING/YOU work.out

Intended: 'Zhangsan does work out.'

b. A: 'Zhangsan doesn't work out.'

B: Zhangsan {bing/you} mei you bu jianshen.

Zhangsan BING/YOU NEG have NEG work.out

'It's not the case that Zhangsan doesn't work out.'

You but not *bing* implies the speaker's impatience with the hearer. For example, the use of *you* in (2) implies that the speaker thinks that it is very obvious that Zhangsan doesn't work out (perhaps he looks very weak), and that the hearer should have known this fact. Because of this impatience implication, *you* can sometimes sound rude and is inappropriate on formal occasions, such as when correcting a superior or a guest. *Bing* does not have this impatience implication.

I will begin my study of *bing* and *you* with a syntactic analysis, and argue that they are located between C and T (section 2). Section 3 will then present the distributional data of *bing* and *you*, namely: *bing* and *you* can be used metalinguistically to correct not at-issue content and even non-linguistic statements; *bing* but not *you* can contradict a part in the speaker's own sentence, is compatible with speaker-oriented adverbs, and can be embedded.

Based on these key differences between *bing* and *you*, I will propose semantic-pragmatic analyses of them in section 4: *bing* presupposes that its prejacent (i.e. $\neg p$) contrasts with a salient proposition, while *you* has the conventional implicature that the speaker believes that

its prejacent (i.e. $\neg p$) was already in the Common Ground (CG). Despite this belief of the speaker, they still reiterate $\neg p$ to remind the hearer of it, deriving *you*'s impatience inference.

Bing and *you* are associated with the illocutionary force, which is usually assumed to reside high in C (e.g. Repp 2013), but syntactically they are located lower, between C and T. To account for this mismatch between their syntactic positions and high illocutionary force, I posit illocutionary force heads $\text{Force}^0_{\text{bing}}$ and $\text{Force}^0_{\text{you}}$, covert operators located in C. They license *bing* and *you* respectively by agreement:

- (4)a. [CP $\text{Force}^0_{\text{bing}}$... *bing* NEG [TP p]]
 b. [CP $\text{Force}^0_{\text{you}}$... *you* NEG [TP p]]

To derive the apparent syntax-semantics mismatch, I suggest that *bing* and *you* are semantically vacuous, and the corrective meanings are actually contributed by $\text{Force}^0_{\text{bing}}$ and $\text{Force}^0_{\text{you}}$ (5)–(6). Negation is interpreted in-situ as logical negation $\neg$, and contrast in (5) is defined in terms of implication of a third proposition provided by the question under discussion, following (Winter and Rimon 1994) and (Toosarvandani 2014).

- (5) $[[\text{Force}^0_{\text{bing}} \text{ NEG } p]]^c(w) = 1$ iff p is false
 Presupposition: $\exists r: \exists q: q$ is salient $\wedge q \Rightarrow r \wedge \neg p \Rightarrow \neg r$
 “There is a salient proposition q and a proposition r supplied by the question under discussion such that q implies r and $\neg p$ implies $\neg r$.”
- (6) $[[\text{Force}^0_{\text{you}} \text{ NEG } p]]^c(w) = 1$ iff p is false
 Conventional implicature: $\forall w' [w' \text{ is compatible with what the speaker}_c \text{ knows in } w \rightarrow \neg p \in \text{CG}_{w'}]$
 “The speaker_c believes that $\neg p$ was in the CG.”

For convenience, the rest of the paper will call the illocutionary force associated with *bing* and *you* simply *bing* and *you* respectively, and not include Force^0 in the analyses, even though the illocutionary force of *bing* and *you* is higher than their syntactic positions. My analysis of *you* has two hallmark properties of evidential markers, making it both a CG manager and an evidential marker. First, *you* encodes the source of evidence a speaker has for a given proposition, and constrains the source to be direct evidence. Second, *you* is speaker-orientated in root assertions.

This paper follows a long tradition in the Chinese literature that studies *bing* and *you* together. Section 5 presents a summary of the previous generalizations and analyses of *bing* and *you* in the Chinese literature and compares them with the current one.

This paper extends beyond *bing* and *you* in Mandarin Chinese with two broad typological comparisons in sections 6 and 7 respectively. Section 6 compares corrective markers (i.e. particles that require a proposition to correct) across languages, further supporting the current analysis that *you*, like the other corrective markers, manages the Common Ground. Section 7 compares *bing* and the coordinator ‘but’, further supporting the view that *bing*, like ‘but’, marks contrast.

Section 6 compares *bing* and *you* with corrective markers *mica* in Italian and *thorī* in Hindi-Urdu. *Mica* and *thorī* appear to behave very similarly to *bing* and *you*: they must co-occur with negation, and also require a salient proposition to correct. The following examples demonstrate this requirement for Italian *mica*:

(7)Context: S and A live in Amherst and want to go to a party in NYC.

A: How are we going to get there?

S: Mia sorella non ha (#mica) la macchina questo fine settimana,
my sister NEG has (#MICA) the car this weekend
altrimenti ci avrebbe accompagnato volentieri.
otherwise us would.have accompanied gladly

‘My sister doesn’t have (#mica) the car this weekend, otherwise she would have gladly given us a ride.
(Frana & Rawlins 2019:6)

(8)A: How are we going to get there? Can your sister give us a ride?

S: Mia sorella non ha (mica) la macchina. Ha 13 anni!

My sister NEG has (MICA) the car. Has 13 years!

‘My sister DOESN’T have a car, she is 13!’
(Frana & Rawlins 2019:6)

There have been two different types of analyses for *mica* and *thori*., which correspond to my analyses of *you* and *bing* respectively. One type analyzed *mica* as a *Common Ground (CG) manager*, in the same vein as my analysis of *you*. For example, Frana & Rawlins (2019) (henceforth F&R) argued that *mica* presupposes that the speaker is sure that *mica*’s prejacent should not be added to the CG. The other type analyzed *mica* as a *marker of mirativity or contrast*, parallel to my analysis of *bing*. For example, Mari (2024) argued that *mica* denies an inference drawn from the evidence based on stereotypes, norms or commonsense reasoning, and Cinque (1976) argued that *mica* denies a prior claim or a salient expectation.

F&R presented evidence that challenges Cinque’s contrast-marker analysis but supports theirs. I will apply F&R’s test for *bing* and *you* and show that it supports my proposal: *bing* is a contrast marker because *bing* behaves exactly like what Cinque predicts *mica* should behave but doesn’t; *you* is a CG manager, parallel to F&R’s analysis of *mica*, but *you*’s contribution to the discourse is slightly different from what they claimed *mica*’s contribution was.

After comparing *bing* and *you* with *mica* and *thori*., section 7 moves on to languages that don’t seem to have a dedicated corrective marker, such as English, French and German. I will argue that despite this appearance, these languages do have a corrective marker—a silent Op(erator) equivalent to Mandarin *bing* that occurs in the first conjunct of ‘not...but...’ sentences (e.g. *John Op doesn’t work out, but goes on long walks*). With this operator, these sentences behave almost identically to *bing*-sentences, including their requirement of syntactic negation. Based on this parallel between *bing* and Op, I suggest that perhaps all corrective markers syntactically select for negation. More broadly, by demonstrating the striking similarities *between bing* and different flavors of ‘but’ in sections 4.1 and 7, I present a unifying analysis of *bing*, the silent Op in ‘not...but...’ sentences, and ‘but’ across languages, and argue that they all mark contrast between two propositions: *bing* and Op contrast their prejacent with a salient proposition, whereas ‘but’ contrasts its two conjuncts.

Section 8 concludes and discusses the implications of the typological analysis. My work here enriches the typology of corrective markers by arguing that they fall into two distinct categories—contrast markers and Common Ground managers. Both these types are attested in Mandarin, and they are more prevalent than we thought: even languages that don’t seem to have a corrective marker do. While all corrective markers require a proposition to correct, they do so for different reasons. This requirement is an inherent part of the presupposition of contrast markers, for example, *bing*’s presupposition that its prejacent should contrast with a salient proposition. In contrast, CG managers require a proposition to correct due to pragmatic

reasoning. For example, *you* has this requirement because unless the hearer has said *p* or given evidence that they believe *p*, it is redundant for the speaker to say $\neg p$, a proposition they believe was already in the CG. It is odd to use *mica* and *thori*: out of the blue to assert that *p* should not be added to the CG, unless someone had said *p*.

2. Syntax of *bing* and *you*

This section argues that *bing* and *you* are located between C and T. Because *bing* and *you* immediately precede negation, subsection 2.1 will first discuss the possible positions for negation in Mandarin. I will show that there are three positions for negation on the clausal spine: what I call Neg₁ is located above C; Neg₂ is between C and T; and Neg₃ between T and V. Then subsection 2.1.4 will show that *bing* and *you* must appear immediately before Neg₂, but not before Neg₁ or Neg₃, putting *bing* and *you* between C and T.

2.1. Three positions for negation in Mandarin

Mandarin has two markers for verbal and predicative negation: *bu* and *mei*. *Mei* is used with *you* ‘have’, which is both the perfective auxiliary and the main verb meaning ‘own’, while *bu* is used elsewhere.

- (9)a. wo {bu/*mei} chi fan.
 I NEG eat rice
 ‘I don’t eat rice.’
- b. wo {*bu/mei} you chi fan.
 I NEG have eat meal
 ‘I haven’t eaten.’
- c. wo {*bu/mei} you qian.
 I NEG have money
 ‘I don’t have money.’

Three pieces of evidence suggest that there are three positions for negation on the clausal spine: Neg₁, in the left-peripheral CP in negative polar questions; Neg₂, above the TP; and Neg₃, above the VP.

2.1.1. Test 1: Co-occurrence of negations

First, these three positions for negation can occur at the same time:

- (10) wo bu mei you bu chi fan ma?
 I NEG₁ NEG₂ have NEG₃ eat meal Q
 ‘Isn’t it the case that it’s not that I don’t eat?’

2.1.2. Test 2: Negation's scope with respect to adverbs

Second, facts involving scope put Neg₂ between C and T, and Neg₃ between T and V. I discuss scope relative to two preverbal adverbs: the habitual adverb *zongshi* 'always' and the speaker-oriented adverb *dagai* 'probably'. Following Cinque's (1999) hierarchy, I assume that *dagai* 'probably' is located above T, while *zongshi* 'always' is between T and V.

Negation can appear before or after *zongshi* 'always', which corresponds to wide and narrow scope of negation respectively, putting Neg₂ above *zongshi*, and Neg₃ below it:

- (11) a. wo zongshi bu chi fan.
 I always NEG₃ eat rice
 'It's always the case that I don't eat rice.'
- b. wo bu zongshi chi fan.
 I NEG₂ always eat rice
 'I don't always eat rice.'

In declaratives, negation can only appear after *dagai* 'probably' but not before it, suggesting that Neg₂ is located below *dagai*:

- (12) a. ta dagai bu chi fan.
 he probably NEG₂ eat rice
 'He probably doesn't eat rice.'
- b. *ta bu dagai chi fan.
 he NEG₁ probably eat rice

2.1.3. Test 3: Two readings of negative polar questions (NPQs)

A negative polar question can check the truth and falsity of a negative proposition (*inner negation reading*) or a positive proposition (*outer negation reading*). For example, (13) has two interpretations: 'Isn't it the case that Zhangsan works out?' (outer negation reading, equivalent to the English question *Doesn't Zhangsan work out?*), a biased question with the speaker's prior bias towards the positive answer 'Zhangsan work out'; or 'Is it the case that Zhangsan doesn't work out?' (inner negation reading, equivalent to the English question *Does Zhangsan not work out?*), where the speaker wants to know the truth and falsity of the proposition 'Zhangsan doesn't work out'.

- (13) Zhangsan bu jianshen ma?
 Zhangsan NEG work.out Q
 ✓Outer negation reading: 'Isn't it the case that Zhangsan works out?'
 ✓Inner negation reading: 'Is it the case that Zhangsan doesn't work out?'

I argue that while the two readings of (13) have the same string of words, they actually correspond to two different positions for negation: the outer negation reading is generated by NPQs with Neg₁, and the inner negation reading is generated by NPQs with Neg₂.

There are three pieces of evidence for this. First, *bu* is pronounced differently in these two readings: *bu* is prosodically unstressed in the outer negation reading but stressed in the inner negation reading, suggesting that Neg₁ and Neg₂ have different phonological realizations. Second, when *bu* precedes the high adverb *dagai* ‘probably’ (leading to Neg₁), it only has the outer negation reading:

- (14) ta bu dagai chi fan le ma?
 he NEG₁ probably eat rice ASP Q
 ‘Isn’t it the case that he probably ate rice?’

When *bu* follows the high adverb *dagai* ‘probably’ (leading to Neg₂), it only has the inner negation reading:

- (15) ta dagai bu chi fan ma?
 he probably NEG₂ eat rice Q
 ‘Is it the case that he probably doesn’t eat rice?’

Finally, when the NPQ includes a negative polarity item (NPI), it only has the inner negation reading, suggesting that only the inner negation reading can license NPIs, but the outer negation reading cannot:

- (16) Zhangsan bu zuo renhe yundong ma?
 Zhangsan NEG₂ do any exercise Q
 Only possible reading (inner negation): ‘Is it the case that Zhangsan doesn’t do any exercise?’

Assuming that *bu* must be in the same clause as the NPI to license it (Wang & Hsieh 1996), this suggests that in the outer negation reading, Neg₁ is outside the TP, but in the inner negation reading, Neg₂ is inside the TP. Following Repp (2013), I take Neg₁ to be an illocutionary operator in the left peripheral CP. Thus, I have diagnosed three different positions for negation in Mandarin: Neg₁ > C > Neg₂ > T > Neg₃ > V.

2.1.4. Syntactic positions of *bing* and *you*

Having argued for three positions for negation, I argue with four tests that *bing* and *you* always appear immediately before Neg₂ (17a), but not before Neg₁ (17b) or Neg₃ (17c). There is debate in the literature about where Neg₂ is exactly: Huang (1988) claimed that it is a bound morpheme in Infl; Ernst (1995) claimed that it is an adverb in Spec, AuxP;¹ Zhou (1999) argued that there is a Neg projection above AspP. But all these proposals located Neg₂ between C and T. I thus assume that *bing* and *you* are also located between C and T.

2.1.5. Test 1: Word order

Bing and *you* always appear immediately before Neg₂ (17a), and may not appear before Neg₁ (17b) or Neg₃ (17c):

¹ Ernst (1995) actually suggested that *bu* in Neg₂ is in Spec, AuxP or Spec, VP. But I think Spec, VP is too low for Neg₂, and more likely for Neg₃ instead.

- (17) a. wo {bing/you} mei you bu chi fan.
 I BING/YOU NEG₂ have NEG₃ eat meal
 ‘It’s not the case that I don’t eat.’
- b. *wo {bing/you} bu chi fan le ma?
 I BING/YOU NEG₁ eat rice ASP Q
- c. *wo mei you {bing/you} bu chi fan.
 I NEG₂ have BING/YOU NEG₃ eat meal

2.1.6. Test 2: Negation’s scope with respect to adverbs

In a declarative sentence with *zongshi* ‘always’, *bing* and *you* must precede the negation with the wide scope, which precedes *zongshi*:

- (18) a. wo {bing/you} bu zongshi chi fan.
 I BING/YOU NEG₂ always eat rice
 ‘I don’t always eat rice.’
- b. *wo zongshi {bing/you} bu chi fan.
 I always BING/YOU NEG₃ eat rice
 Intended: ‘It’s always the case that I don’t eat rice.’

2.1.7. Test 3: Negative polar questions (NPQs)

Bing but not *you* can appear in NPQs (a fact to be discussed later in section 6.2). NPQs that include *bing* only have the inner negation reading:

- (19) Zhangsan bing bu jianshen ma?
 Zhangsan BING NEG₂ work.out Q
 *Outer negation reading: ‘Isn’t it the case that Zhangsan works out?’
 ✓Inner negation reading: ‘Is it the case that Zhangsan doesn’t work out?’

The negation in (19) is prosodically stressed, suggesting that it is Neg₂. When *bing* appears in NPQs, it has to follow high adverbs like *dagai* ‘probably’ rather than precede them:

- (20) a. Zhangsan dagai bing bu jianshen ma?
 Zhangsan probably BING NEG₂ work.out Q
 ‘Is it the case that Zhangsan probably doesn’t work out?’
- b. *Zhangsan bing bu dagai jianshen ma?
 Zhangsan BING NEG₁ probably work.out Q

2.1.8. Test 4: NPI-licensing

Neg₂ can license NPIs in the same clause. This still holds in *bing*- and *you*-sentences:

- (21) a. wo (bing/you) bu taoyan renhe ren.
 I BING/YOU NEG₂ hate any person
 ‘I don’t hate anyone.’
- b. wo (bing/you) mei chi renhe dongxi.
 I BING/YOU NEG₂ eat any thing
 ‘I didn’t eat anything.’

NPs that include *bing* can license NPIs, suggesting that the negation is Neg₂:

- (22) Zhangsan bing bu zuo renhe yundong ma?
 Zhangsan BING NEG₂ do any exercise Q
 Only possible reading (inner negation): ‘Is it the case that Zhangsan doesn’t do any exercise?’

Thus, this section has shown that *bing* and *you* require Neg₂, and are located between C and T.

3. Distribution of *bing* and *you*

Having discussed the syntactic positions of *bing* and *you*, this section discusses their distribution. I will show that *bing* and *you* can be used for metalinguistic correction (subsection 3.1). *Bing* but not *you* can contradict a part in the speaker’s own sentence (subsection 3.2). *You* is incompatible with any speaker-oriented adverb, while *bing* is only incompatible with the high confidence adverb *yiding* ‘definitely’ (subsection 3.3). *You* cannot be embedded at all, while *bing* can be embedded in polar questions, finite complement and adverbial clauses (subsection 3.4).

3.1. Metalinguistic use

Most of the corrective examples we have seen so far are semantic correction, but the negation associated with *bing* and *you* can also be metalinguistic. Metalinguistic negation can correct pragmatic effects such as implicatures (23a), presuppositions (23b), phonology (23c), morphological make-up (23d) and sociolinguistic use (23e).

- (23) a. Chris managed to solve not some of the problems, but all of them.
 b. John hasn’t stopped smoking, he didn’t smoke at all before.
 c. I ate not tom[eiDouz] but tom[a:touz].
 d. I managed to trap not two mongeese but two mongooses.
 e. Grandpa is feeling not lousy but just a tad indisposed. (Based on Horn 2001:370-371)

Bing and *you* can be used metalinguistically. They can correct implicatures (24a), presuppositions (24b), phonology ((24c), which corrects the tone on the first vowel of *fenweigan* ‘vibe’), morphological make-up ((24d), which corrects the reduplicated form of *zixi* ‘carefully’) and sociolinguistic use ((24d), which corrects the choice of words for ‘tomato’).

- (24) a. A: ‘Zhangsan solved some of the problems.’
 B: Zhangsan jie jue de {bing/you} bu shi yixie wenti,

Zhangsan solve DE BING/YOU NEG COP some problem
 er shi suoyou wenti.
 but COP all problem
 ‘What Zhangsan solved was not some of the questions, but all of the questions.’

- b. A: ‘Zhangsan has stopped smoking.’
 B: ta yiqian bing/you bu chouyan.
 he before BING/YOU NEG smoke
 ‘He didn’t smoke before.’
- c. A: Lisi shi fènweigan meinv.
 Lisi COP vibe beauty
 ‘Lisi is a woman who vibes beauty.’
 B: Lisi {bing/you} bu shi fènweigan meinv. Ta shi fènweigan meinv.
 Lisi BING/YOU NEG COP vibe beauty she COP vibe beauty
 ‘Lisi is not *fènweigan* beauty. She is *fènweigan* beauty.’
- d. A: Zhangsan hen qinfen, meitian zixi zixi de gongzuo.
 Zhangsan very diligent every.day carefullycarefullyDE work
 Intended: ‘Zhang is very diligent, and works carefully every day.’
 B: Zhangsan {bing/you} bu shi zixi zixi de gongzuo,
 Zhangsan BING/YOU NEG COP carefullycarefullyDE work
 er shi zizixixi de gongzuo.
 but COP carefully DE work
 ‘Zhangsan doesn’t work *zizixixi*, but works *zizixixi*.’
- e. A: ‘Zhangsan ate some *fanqie* “tomatoes”.’
 B: Zhangsan chi de {bing/you} bu shi fanqie, er shi xihongshi.
 Zhangsan eat DE BING/YOU NEG COP tomato but COP tomato
 ‘What Zhangsan ate was not *fanqie* “tomatoes”, but *xihongshi* “tomatoes”.’

Bing and *you* can even correct non-linguistic statements:

- (25) Context: A puppy approaches Zhangsan, who looks scared. I tell Zhangsan:
 bie pa. ta {bing/you} bu hui yao ni.
 don’t afraid it BING/YOU NEG will bite you
 ‘Don’t be afraid. It won’t bite you.’

3.2. Contradicting not an interlocutor, but a part of the speaker’s own sentence

Example (2) showed that *bing* and *you* can contradict the interlocutor in a dialogue. In addition, *bing* can also contradict the expectation created in the same sentence, and the clause that creates the expectation may precede or follow the *bing*-clause. (26)–(27) can be said out-of-the-blue

to ensure that the proposition to be corrected comes from the speaker themselves rather than the hearer. This contrasts with *you*, which cannot contradict a part in the same sentence:²

- (26) Context: You ask me to tell you something about my friend Zhangsan, who you don't know anything about. I say:

Zhangsan meitian qu jianshenfang, danshi ta {bing/#you} bu jianshen.
 Zhangsan every.day go gym but he BING/YOU NEG work.out
 'Zhangsan goes to the gym every day, but he doesn't work out.'

- (27) Context: You ask me to tell you something about my friend Zhangsan, who you don't know anything about. I say:

Zhangsan {bing/#you} bu jianshen, danshi que hen jiankang.
 Zhangsan BING/YOU NEG work.out but PART very healthy
 'Zhangsan doesn't work out, but he is very healthy.'

3.3. Compatibility with speaker-oriented adverbs

You cannot co-occur with any speaker-oriented adverb such as *yiding* 'definitely', *dagai* 'probably', *duoban* 'more than likely' and *keneng* 'possibly':

- (28) A: 'Zhangsan works out.'

B: #Zhangsan {yiding / dagai / duoban / keneng} you bu jianshen.
 Zhangsan {definitely / probably/ more than likely / possibly} YOU NEG work.out
 Intended: 'Zhangsan {definitely / likely/ probably / possibly} doesn't work out.'

Bing can co-occur with all these speaker-oriented adverbs except the high confidence one *yiding* 'definitely':

- (29) A: 'Zhangsan works out.'

B: Zhangsan {*yiding / dagai / duoban / keneng} bing bu jianshen.
 Zhangsan {definitely / probably/ more than likely / possibly} BING NEG work.out
 'Zhangsan {*definitely / likely/ probably / possibly} doesn't work out.'

² (26) is actually possible with *you*, but I argue with two pieces of evidence that the possible *you* in (26) means 'at the same time', and is distinct from the *you* studied in this paper. First, while the *you* and *bing* studied in this paper require negation as we saw in (2), *you* 'at the same time' doesn't:

- (i) Context: You ask me to tell you something about my friend Zhangsan, who you don't know anything about. I say:

Zhangsan meitian qu jianshenfang, danshi ta {#bing/you} hen ai chi laji shipin
 Zhangsan every.day go gym but he BING/YOU very love eat junk food
 'Zhangsan goes to the gym every day, but at the same time he loves junk food.'

The second piece of evidence that the two *yous* are different is that *you* 'at the same time' does not have the obviousness / impatience implication that the *you* studied in this paper has. (26) and (i) do not imply that it is obvious that Zhangsan doesn't work out or it is obvious that he loves junk food.

Thus, I argue that *you* is polysemous. This paper focuses on its corrective use, which can only be used in a dialogue. In addition to correction and 'at the same time', *you* can also conjoin predicates, and can mean 'again'. *Bing* is also polysemous and can conjoin predicates. These other uses will be discussed at the end of section 4.

- (30) Zhangsan meitian qu jianshenfang, danshi ta
 Zhangsan every.day go gym but he
 {*yiding / dagai / duoban / keneng} bing bu jianshen.
 {definitely / probably / more than likely / possibly} BING NEG work.out
 ‘Zhangsan goes to the gym every day, but he { *definitely / likely / probably / possibly }
 doesn’t work out.’

3.4. Embeddability

You cannot be embedded at all, while *bing* can be embedded in polar questions, finite complement and adverbial clauses.

3.4.1. *Wh*-questions

Bing and *you* cannot appear in *wh*-questions:³

- (31) #Zhangsan {bing/you} bu zai nali jianshen?
 Zhangsan BING/YOU NEG at where work.out
 Intended: ‘What’s the location x such that Zhangsan doesn’t work out at x?’

3.4.2. Polar questions

You cannot appear in polar questions, but *bing* can, a fact to be discussed in depth in section 6.2:

- (32) Zhangsan bing bu jianshen ma?
 Zhangsan BING NEG work.out Q
 Inner negation reading only: ‘Is it the case that Zhangsan doesn’t work out?’

- (33) #Zhangsan you bu jianshen ma?
 Zhangsan YOU NEG work.out Q

3.4.3. Imperatives

Bing and *you* cannot appear in imperatives:

- (34) #{bing/you} bu xu jianshen.
 BING/YOU NEG allowed work.out
 Intended: ‘Don’t work out.’

3.4.4. Conditionals

You cannot appear in the antecedent of conditionals, but *bing* can:

³ *Bing* may only be possible in (31) if it is interpreted as an echo question.

- (35) A: 'I hope Zhangsan will come to the party.'
 B: ruguo Zhangsan {bing/#you} bu hui lai paidui, name jiu
 if Zhangsan BING/YOU NEG will come party then PART
 mei you ban de biyao le.
 NEG have host DE necessity LE
 'If Zhangsan won't come to the party, then there's no point in hosting it.'

Both *bing* and *you* can appear in the consequent of conditionals, though *you* is slightly marked:

- (36) A: 'Zhangsan has a chance of having diabetes in the future.'
 B: ruguo Zhangsan meitian jianshen,
 if Zhangsan every.day work.out
 name ta {bing/?you} bu hui de tangniaobing.
 then he BING/YOU NEG will get diabetes
 'If Zhangsan works out every day, then he won't have diabetes.'

3.4.5. 'Because'-clauses

Bing can appear in *yinwei*- 'because'-clauses, while *you* cannot:

- (37) yinwei Zhangsan {bing/#you} bu jianshen, suoyi ta rongyi pilao.
 because Zhangsan BING/YOU NEG work.out so he easily tired
 'Because Zhangsan doesn't work out, he gets tired easily.'

3.4.6. 'Now-that'-clauses

Bing can appear in *jiran*- 'now that'- (or 'given that'-) clauses, while *you* cannot:

- (38) jiran Zhangsan {bing/#you} bu jianshen,
 now.that Zhangsan BING/YOU NEG work.out
 name ta bu xuyao jianshenfang huiyuan.
 then he NEG need gym membership
 'Now that Zhangsan doesn't work out, he doesn't need a gym membership.'

3.4.7. Relative clauses

You cannot appear in relative clauses, but *bing* can:

- (39) zhege jingjiren {?bing/#you} mei you kanshang de geshou
 this agent BING/YOU NEG have spot DE singer
 houlai que cheng le da mingxing.
 later PART bome LE big star
 'The singer that this agent failed to spot later became a big celebrity.'

3.4.8. Complement clauses

Complement clauses in Mandarin Chinese vary in finiteness. The complement of factive and non-factive verbs is a finite clause. *Bing* can appear in the complement clause of both factive verbs like ‘realize’ and ‘surprised’ and non-factive verbs like ‘say’ and ‘think’, while *you* cannot appear in any complement clause:⁴

- (40) Lisi {shuo/renwei/hen jingya/yishidao} Zhangsan bing bu jianshen.
 Lisi {say/think/very surprised/realize} Zhangsan BING NEG work.out
 ‘Lisi {said/thought/was surprised/realized} that Zhangsan doesn’t work out.’
- (41) #Lisi {shuo/renwei/hen jingya/yishidao} Zhangsan you bu jianshen.
 Lisi {say/think/very surprised/realize} Zhangsan YOU NEG work.out

The complement of *xihuan* ‘like’ has been argued to be a vP (Grano 2015) or nonfinite clause (Ussery et al. 2016) because it lacks the properties of a full clause, for example it doesn’t allow overt subjects. Neither *bing* nor *you* can appear in the complement of *xihuan* ‘like’, suggesting that they can only occur in full clauses. This is expected if they are located between C and T, and the complement of *xihuan* lacks TP and CP.

- (42) *Zhangsan xihuan {bing/you} bu chuan wazi.
 Zhangsan like BING/YOU NEG wear socks
 Intended: ‘Zhangsan prefers not wearing socks.’

The following table summarizes the distribution of *bing* and *you* in this section:

	<i>Bing</i>	<i>You</i>
Metalinguistic negation	✓	✓
Contradict a part of the speaker’s own sentence	✓	*
Compatible with speaker-oriented adverbs	✓ (except <i>yiding</i> ‘definitely’)	*
Embeddable in <i>wh</i> -questions	*	*
Embeddable in polar questions	✓	*
Embeddable in imperatives	*	*
Embeddable in the antecedent of conditionals	✓	*
Embeddable in the consequent of conditionals	✓	✓
Embeddable in ‘because’-clauses	✓	*
Embeddable in ‘now that’-clauses	✓	*
Embeddable in relative clauses	✓	*
Embeddable in finite complement clauses	✓	*
Embeddable in nonfinite complement clauses	*	*

Table 1: Distribution of *bing* and *you*.

⁴ *You* may appear to be allowed in the complement of *shuo* ‘say’, but the complement has to be interpreted as a quote of Lisi.

4. Semantics-pragmatics of *bing* and *you*

Having examined the syntax and distribution of *bing* and *you*, this section focuses on their semantics. The following two subsections will present the semantic-pragmatic analyses of *bing* and *you* respectively, and argue that *bing* requires that its prejacent contrast with a salient proposition, while *you* implies that its prejacent was already in the Common Ground. Subsection 4.3 will then argue that *bing* and *you* contribute not at-issue content, specifically: *bing*'s contribution is presuppositional, while *you*'s is a conventional implicature.

4.1. *Bing*

This subsection will study the meaning of *bing* by comparing it with a word whose meaning has been well-studied in the literature—*but* in English. I will show that English *but* and *bing* have very similar uses, and then adapt the semantic-pragmatic proposals for *but* for *bing*.

First I present some background on *but* in English. *But* has at least three uses: counterexpectation, semantic opposition and correction (e.g., Toosarvandani's 2014 taxonomy). Each use of *but* in English requires contrast of some sort. In the counterexpectational use, the first conjunct creates an expectation that is rejected by the second conjunct (43). In contrast, neither of the other two uses necessarily creates an expectation. In semantic opposition, the conjuncts contrast with each other in two positions (44). Corrective *but* requires presence of negation in the first conjunct and absence of negation in the second conjunct (45).

- | | |
|---|----------------------------|
| (43) Zhangsan looks fit, but he doesn't work out. | <i>counterexpectation</i> |
| (44) Zhangsan is tall but Lisi is short. | <i>semantic opposition</i> |
| (45) Zhangsan doesn't work out, but goes on long walks. | <i>correction</i> |

I will argue that *bing*-sentences parallel counterexpectational and corrective *but* sentences in many ways. The rest of this subsection will argue for the following parallel between *bing* and counterexpectational *but*: the second conjunct of counterexpectational *but* rejects the expectation created by the first conjunct, in the same way that the *bing*-sentence rejects the expectation created by another clause. In fact, *bing* can generally occur in counterexpectational sentences:

- (46) A: 'Zhangsan looks fit.'
B: (danshi) Zhangsan bing bu jianshen
but Zhangsan BING NEG work.out
'Zhangsan doesn't work out.'
- (47) Zhangsan kanshangquhen jiankang, danshi ta qishi bing bu jianshen.
Zhangsan look very fit but he actually BING NEG work.out
'Zhangsan looks fit, but he actually doesn't work out.'

Besides these similarities, I will also observe some small differences between *bing* and counterexpectational *but*. Based on these observations, I will introduce Winter & Rimon's (1994) and Toosarvandani's (2014) analysis of counterexpectational *but*, and then adapt them for *bing*. This analysis is based on implication and question under discussion (QUD):

(48) *Meaning of bing not p*

QUD = *r*?

$\llbracket \text{bing not } p \rrbracket^c(w) = 1$ iff *p* is false

Presupposition: $\exists r: \exists q: q$ is salient $\wedge q \Rightarrow r \wedge \neg p \Rightarrow \neg r$

“There is a salient proposition *q* such that *q* implies *r* and $\neg p$ implies $\neg r$.”

After showing the parallel between counterexpectational *but* sentences and *bing*-sentences in this subsection, section 7 will show that corrective *but* sentences are even more similar to *bing*-sentences, but what parallels *bing not p* is not corrective *but* itself, but the little-noticed negation in the first conjunct of corrective *but* sentences.

4.1.1. *Bing not p* is licensed if *p* is implied by a salient proposition

The examples so far involving *bing* seemed to suggest that *bing not p* requires *p* to be salient:

(49) *Meaning of bing not p (Version 1)*

$\llbracket \text{bing not } p \rrbracket^c(w) = 1$ iff *p* is false

Defined only if *p* is salient

“*p* is salient.”

By salient, I mean having been either mentioned explicitly or implied by nonlinguistic statements. For example, as we saw in (2), repeated below, the sentence *Zhangsan bing not works out* (*bing not p*) is infelicitous out of the blue, but is fine if the interlocutor had said *Zhangsan works out* (*p*) explicitly. In the non-linguistic example in (25), *the puppy bing won't bite Zhangsan* is licensed because *the puppy will bite Zhangsan* is implied by Zhangsan's fear.

(2)A: ‘Zhangsan works out.’

B: Zhangsan {bing/you} *(bu) jianshen.

Zhangsan BING/YOU NEG work.out

‘Zhangsan doesn't work out.’

This and the next subsection will show that *bing* can be licensed even if *p* itself is not salient, but instead something weaker than *p* is salient, suggesting that *bing*'s requirement is more relaxed than requiring *p* to be salient. Contrast (2) with (47) and (46), repeated below, where *bing* can be licensed in denial of the proposition ‘Zhangsan looks fit’ (*q*):

(46) A: ‘Zhangsan looks fit.’

B: (danshi) Zhangsan bing bu jianshen

but Zhangsan BING NEG work.out

‘Zhangsan doesn't work out.’

This parallels the use of English counterexpectational *but* in (43). How can ‘Zhangsan doesn't work out’ contrast with ‘Zhangsan looks fit’? Winter & Rimon's (1994) idea was that what is being contrasted is not ‘Zhangsan looks fit’ itself, but a possible consequence of it. To formalize this and account for (43), Winter & Rimon introduced the idea of implication in a possible world, which Toosarvandani (2014) developed based on modal semantics. They claimed that in (43), *Zhangsan looks fit* (*q*) entails *Zhangsan works out* (*p*) in the most normal

worlds, which they called *q implies p* ($q \Rightarrow p$). Toosarvandani formalized $q \Rightarrow p$ with a necessity modal expressing universal quantification over possible worlds, and this modal is relativized to an epistemic modal base and a stereotypical ordering source. In plain words, *Zhangsan looks fit $\Rightarrow$ Zhang works out* is true iff in all the epistemically accessible worlds that best correspond to the normal course of events, Zhangsan looks fit entails that Zhangsan works out. Thus, they claimed that counterexpectational *but* requires its first conjunct to imply the negation of the second conjunct.

Applying this analysis to the *bing*-sentences (46) and (47), there the salient *q* ('Zhangsan looks fit') is not identical to *p* ('Zhangsan works out'), but rather *q* implies *p* ($q \Rightarrow p$). Their felicity thus suggests that we need to relax *bing not p*'s presupposition: rather than requiring *p* to be salient, *bing not p* only requires *p* to be *implied* by a salient proposition *q* (50). The direct denial in (2) is a special case where *q* is identical to *p*, and therefore *q* implies *p*.

(50) *Meaning of bing not p (Version 2)*

$\llbracket \text{bing not } p \rrbracket^c(w) = 1$ iff *p* is false

Defined only if $\exists q$: *q* is salient $\wedge q \Rightarrow p$

"There is a salient proposition *q* such that *q* implies *p*."

4.1.2. *Bing not p* is licensed if *not p* implies not *r*, and a salient proposition *q* implies *r*

The dialogue in (51) suggests that even this already relaxed presupposition is too strict: 'I'm hungry' (*q*) doesn't normally entail that the restaurants are open (*p*) because the restaurants' opening hours do not normally depend on speaker A's hunger status, but *bing* is still fine here:

(51) A: 'I'm hungry.'

B: (danshi) fandian bing mei you kai
 but restaurant BING NEG have open
 'The restaurants aren't open.'

In parallel, English counterexpectational *but* can coordinate this *p* and this *q*:

(52) I'm hungry, but the restaurants aren't open.

Winter & Rimon (1994) accounted for (52) by introducing a third proposition *r* (i.e. I will get to eat) that stands in an implicational relationship to *p* and *q*. They suggested that counterexpectational *but* requires its first conjunct $q \Rightarrow r$ (i.e. in the most normal worlds, I am hungry entails that I will get to eat), and its second conjunct $\neg p \Rightarrow \neg r$ (i.e. in the most normal worlds, the restaurants aren't open entails that I will not get to eat). Following Winter & Rimon (1994), I call any two propositions $\neg p$ and *q* that stand in this implicational relationship with a third proposition *r* *contrast*. Figure 2 illustrates a pair of contrasts $\neg p$ and *q*:

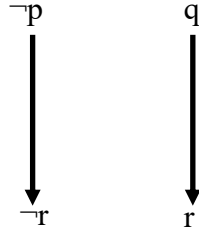


Figure 2: A pair of contrasts $\neg p$ and q .

Another, perhaps more intuitive way is to think of a pair of contrasts to be “counterarguments” to each other. We may think of q entails r as q being an “argument” for r because if q is true, then r is necessarily true. Then q implies r may be considered as q being a slightly weaker argument for r because in the most normal worlds, if q is true, then r is also true. This is a weaker argument than q entails r because the exceptions where this argument doesn’t hold are if things do not proceed in the most normal way.

We can thus think of contrasted propositions as “arguments” on different sides of a “debate”. In Figure 2, we can think of the debate to be about whether r is true. q and $\neg p$ are contrasted propositions in that q is an argument for r , while $\neg p$ is an argument for $\neg r$.

Based on this idea of contrast / counterarguments, counterexpectational *but* requires its conjuncts to contrast with each other, or to be counterarguments to each other. Applying this analysis to *bing*, I say *bing not p* requires $\neg p$ and a salient proposition q to be a pair of contrasts (48). Using the metaphor of a debate, the user of *bing not p* engages in a debate about whether r is true: by arguing against a salient proposition q ’s argument for r , the speaker uses $\neg p$ to argue for $\neg r$.

(48) *Meaning of bing not p (Version 3)*

$\llbracket \text{bing not } p \rrbracket^c(w) = 1$ iff p is false

Defined only if $\exists q: q$ is salient $\wedge q \Rightarrow r \wedge \neg p \Rightarrow \neg r$

“There is a salient proposition q such that q implies r and $\neg p$ implies $\neg r$.”

One way to argue for $\neg r$ is to simply assert $\neg r$. (46) was such a case, where $\neg p$ (‘Zhangsan doesn’t work out’) is identical to $\neg r$, and thus $\neg p$ implies $\neg r$ trivially, and contrasts with the salient proposition q (‘Zhangsan looks fit’), which implies p/r (‘Zhang works out’). In direct denials like (2), the hearer and the listener argue for r and $\neg r$ by directly asserting it respectively: (2A) argues for r by asserting r , while (2B) argues for $\neg r$ by asserting $\neg r$.

We can consider r , the proposition under debate, to be supplied by the question under discussion (QUD). There is a tradition that does so in the literature on *but*. For example, Umbach (2004, 2005) claimed that when answering a question, one conjunct of a *but*-sentence must confirm the question, while the other conjunct must deny it. Following Roberts’ (1996) theory that any sentence in a coherent discourse is the answer to an explicit or implicit question, Umbach claimed that a *but*-sentence gives a ‘yes but no’ answer to the QUD. Toosarvandani (2014) incorporated both the QUD theory and Winter & Rimon’s (1994) contrast in his analysis,

and proposed that counterexpectational *but* requires its conjuncts to answer the polar QUD $r?$, such that the first conjunct of *but* implies r , and the second conjunct of *but* implies $\neg r$ (53).⁵

(53) *Toosarvandani's (2014) proposed meaning of q but p*

QUD = $r?$

$\llbracket q \text{ but } p \rrbracket = \llbracket q \rrbracket \wedge \llbracket p \rrbracket$

Presupposition: $q \Rightarrow r \wedge p \Rightarrow \neg r$

Following Umbach and Toosarvandani, I consider r , the claim under debate in the *bing*-sentence, to be provided by the QUD $r?$:

(48) *Meaning of $bing$ not p (Final Version)*

QUD = $r?$

$\llbracket bing \text{ not } p \rrbracket^c(w) = 1$ iff p is false

Defined only if $\exists q: q$ is salient $\wedge q \Rightarrow r \wedge \neg p \Rightarrow \neg r$

“There is a salient proposition q such that q implies r and $\neg p$ implies $\neg r$.”

For example, we can think of the interlocutors in (51) to be debating about the answer to this polar QUD: *Will speaker A get to eat?* (51A) is an argument for the *yes*-answer to this question, while (51B) is an argument for the *no*-answer.

4.1.3. Difference between *bing* and English counterexpectational *but*

My analysis of *bing*, where the *bing*-sentence contrasts with a salient proposition, is adapted from Winter & Rimon's (1994) and Toosarvandani's (2014) analysis of counterexpectational *but*, where the second conjunct contrasts with the first conjunct. There is a difference between this analysis of counterexpectational *but* and my analysis of *bing*: while *but* requires its prejacent to contrast with the first conjunct q , *bing* only requires its prejacent to contrast with *a salient proposition*. In other words, the proposition that contrasts with *but*'s prejacent is syntactically constrained—it has to be the first conjunct of *but*⁶—whereas the proposition that

⁵ This was not actually Toosarvandani's (2014) proposal. For Toosarvandani, it would suffice if r is a proposition contained in the QUD. I have adapted his proposal to specify that the QUD itself is $r?$ in anticipation of my proposal for *bing* and corrective *but* because *bing*- and corrective *but* sentences answer a simple polar question $r?$.

⁶ Counterexpectational *but* can be used to contrast with the hearer:

(i) A: I'm hungry.

B: But the restaurants aren't open.

Winter and Rimon (1994) and Toosarvandani (2014) did not discuss use of *but* in dialogues like this, but we can imagine that they may say that there is a covert first conjunct in (iB) (*Yes,*) *but the restaurants aren't open* that has the effect of copying the hearer's statement: (*You are hungry,*) *but the restaurants aren't open*. This analysis is plausible because counterexpectational *but* is odd in direct denial of the hearer, in contrast to Mandarin *bing*:

(ii) A: Zhangsan works out.

B: ??But he doesn't.

The correction in (iiB) does not sound very cooperative, but the use of *bing* in (2) does not have this connotation. We can make sense of this contrast if we say there is a covert first conjunct in (iiB) that is copied from the hearer:

contrasts with *bing*'s prejacent is freer and only has to be salient. This predicts that *bing* may have a wider and freer distribution than English *but*, and this subsection will show that this prediction is borne out. (54a-b) use verbs 'lie' and 'frame', which suggest that the speaker believes the propositions 'Zhangsan went to work out' and 'Lisi was corrupt' are false:

- (54) a. Zhangsan sahuang shuo [_q ziji jianshen], qishi [_p ta bing bu jianshen].
 Zhangsan lie say self work.out actually he BING NEG work.out
 'Zhangsan lied that he works out, actually he doesn't.'
 b. Zhangsan zaizang [_q Lisi tanwu], qishi [_p ta bing mei you].
 Zhangsan frame Lisi corrupt actually she BING NEG have
 'Zhangsan framed Lisi for corruption, actually she wasn't corrupt.'

Examples (54a-b) are fine because the propositions that contrast with *bing*'s prejacent are the clauses embedded under the verbs 'lie' and 'frame'—'Zhangsan worked out' and 'Lisi was corrupt'. Crucially, the contrasted propositions cannot be the entire first conjuncts because in the most normal worlds, 'Zhangsan lied that he went to work out' does not entail that Zhangsan went to work out, but the contrary. Likewise, in the most normal worlds, 'Zhangsan framed Lisi for corruption' does not entail that Lisi was corrupt, but the contrary.

In contrast, following Winter and Rimon's and Toosarvandani's analysis that the proposition that contrasts with *but*'s prejacent has to be its syntactic first conjunct *q*, we expect the English counterparts of (54a-b) with *but* to be bad. This is borne out, as the presence of *but* in the following examples is worse than its absence:

- (55) a. [_q Zhangsan lied that he went to work out], (??but) [_p actually he stayed at home].
 b. [_q Zhangsan framed Lisi for corruption], (#but) [_p actually she was honest].

4.2. *You*

Bing and *you* have an important but subtle difference in meaning: unlike *bing*, the use of *you* in *you not p* implies that the speaker considers *not p* to be very obvious, and thus signals the speaker's impatience with the hearer. Consider (25), repeated below. Here the use of *you* implies the speaker thinks Zhangsan is too cowardly and should not fear a puppy. In contrast, *bing* does not have this implication.

(25) Context: A puppy approaches Zhangsan, who looks scared. I tell Zhangsan:

bie pa. ta {bing/you} bu hui yao ni
 don't afraid it BING/YOU NEG will bite you
 'Don't be afraid. It won't bite you.'

Since propositions already in the Common Ground seem obvious to the speaker, I therefore propose that *you* presupposes that the speaker believes $\neg p$ was already in the CG (56). Despite that, the speaker still repeats $\neg p$ to remind the hearer of it, creating the impatience implication.

Zhangsan works out, but he doesn't. This is odd because it is a contradiction, unless *works out* in the two conjuncts has different nuanced interpretations.

- (56) $\llbracket \text{you not } p \rrbracket^c(w) = 1$ iff p is false
 Defined only if $\forall w'$ [w' is compatible with what the speaker_c knows in $w \rightarrow \neg p \in \text{CG}_{w'}$]
 “The speaker_c believes that $\neg p$ was in the CG.”

You is justified in (25) because normally a puppy doesn’t bite. But if the animal in the situation is a hungry wild lion in a savannah, then the use of *you* is very odd because it is odd to presuppose that lions don’t bite:

- (57) Context: A hungry wild lion approaches Zhangsan in a savannah, who is alone and scared.
 I assure Zhangsan on the phone:
 bie pa. ta {bing/#you} bu hui yao ni
 don’t afraid it BING/YOU NEG will bite you
 ‘Don’t be afraid. It won’t bite you.’

In contrast, *bing* is fine in both situations because Zhangsan’s fear makes salient the proposition that the puppy / the lion will bite him.

Instead of following my proposal, one might account for (25) and (57) with the alternative hypothesis that *you* requires its prejacent to be plausible. According to this alternative hypothesis, *you not p* is felicitous in (25) because $\neg p$ (i.e. the puppy won’t bite Zhangsan) is plausible, whereas *you not p* is infelicitous in (57) because $\neg p$ (i.e. the lion won’t bite Zhangsan) is implausible. The following examples challenge this alternative proposal. *You*’s prejacent in (58) is the same as in (25) (i.e. the lion won’t bite Zhangsan) and generally implausible. But the context of (58) makes it clear that the speaker believes that $\neg p$, a generally little-known fact, is nevertheless known to both the interlocutors. The use of *you* is much better in (58) than (25).

- (58) Context: A hungry wild lion approaches Zhangsan in a savannah. Zhangsan and I both grew up in the savannah and are skilled at training lions. I think we both know that despite its scary appearance, the lion wouldn’t actually bite Zhangsan. But Zhangsan is still scared and calls me on the phone. I tell him:
 bie pa. ta {bing/you} bu hui yao ni
 don’t afraid it BING/YOU NEG will bite you
 ‘Don’t be afraid. It won’t bite you.’

Example (59) presents a situation in the reverse, where $\neg p$ is a well-known fact, but the hearer happens to not know it, and the speaker knows that. Here the speaker’s use of *you* is very rude because the speaker should not think that $\neg p$ was in the CG for both the interlocutors.

- (59) Context: I am teaching Xiaoming, a five-year-old, in his first arithmetic lesson.
 Xiaoming: ‘One plus one is three.’
 A: yi jia yi {bing/#you} bu dengyu san.
 one plus one BING/YOU NEG equal three
 ‘One plus one is not three.’

But if the student in this situation is a college student, *you* is justified and signals the speaker’s impatience. *Bing* is fine in (58) and (59) because Zhangsan’s fear and Xiaoming’s statement have made p salient (i.e. respectively, the lion will bite Zhangsan, and one plus one is three).

4.3. *Bing* and *you*'s contributions are not at-issue

So far, I have argued that the illocutionary force of *bing* requires that its prejacent contrasts with a salient proposition, while the illocutionary force of *you* implies that the speaker believes that its prejacent was already in the CG. This subsection argues that their meaning contributions are not at-issue, specifically, *bing* contributes a presupposition, while *you* a conventional implicature.

Bing's requirement of contrast to a salient proposition is not part of *bing*'s assertion because it projects out of environments with negation (60) and modal adverbs like 'probably' (61).

- (60) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
#bu neng shuo Zhangsan bing bu jianshen.
NEG can say Zhangsan BING NEG work.out
Intended: 'We can't say that Zhangsan doesn't work out.'

- (61) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
#Zhangsan dagai bing bu jianshen.
Zhangsan probably BING NEG work.out
Intended: 'Zhangsan probably doesn't work out.'

You cannot be embedded under negation and modals, but I use another test for (non) at-issueness, the challengeability test, to show that its contribution is also not at-issue. Usually, only at-issue content can be accepted or rejected by a hearer (Faller 2002; Papafragou 2006; Murray 2017). Zhangsan's denial in (62) can only target the claim that the puppy won't bite Zhangsan, but not the obviousness of this claim in the case of *you*, or the salience of the claim that the puppy will bite Zhangsan in the case of *bing*.

- (62) Context: A puppy approaches Zhangsan, who looks scared. I tell Zhangsan:
bie pa. ta {bing/you} bu hui yao ni.
don't afraid it BING/YOU NEG will bite you
'Don't be afraid. It won't bite you.'
Zhangsan: wo bu tongyi ni shuo de.
I NEG agree you say DE
'I disagree with what you said.'

Therefore, I assume that *bing not p* and *you not p* have the same assertion as *not p*; *bing* and *you*'s contributions are not at-issue, making them presuppositions, conventional implicatures or conversational implicatures. Their meanings are not cancellable, suggesting they are not conventional implicatures:

- (63) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
#Zhangsan bing bu jianshen. women mei shuo ta jianshen.
Zhangsan BING NEG work.out we NEG say he work.out
Intended: 'Zhangsan doesn't work out. We didn't say he works out.'
- (64) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
#Zhangsan you bu jianshen. women mei you renwei ta bu jianshen.

Zhangsan YOU NEG work.out we NEG have think he NEG work.out
Intended: 'Zhangsan doesn't work out. We didn't think he doesn't work out.'

I claim that *bing* contributes a presupposition, consistent with the view of most Chinese scholars (e.g. Hu 2008; Peng 1999; Wei 2020; Wen 2009; Yang 2008; Yuan & Liu 2025; Yuan 2023); in contrast, *you* contributes a conventional implicature, consistent with Biq (1988) and Yuan & Liu (2025), but not Peng (1999) and Wu (1999), who considered it presuppositional. I provide two arguments for my claim.

First, conventional implicatures cannot be backgrounded, whereas presuppositions can (Potts 2015). Generally, presupposed content can be said explicitly before using the presuppositional trigger (e.g. *John once smoked, but he quit smoking*). It is fine to do so with *bing* but not with *you*:

- (65) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
suiran you ren renwei Zhangsan jianshen,
although have people think Zhangsan work.out
dan qishi ta bing bu jianshen.
but actually he BING NEG work.out
'While some people think that Zhangsan works out, but actually he doesn't.'

- (66) A: 'Zhangsan works out.'
B: ni zai shuo shenme?! zhongsuozhouzhi Zhangsan bu jianshen,
you PROG say what everyone.knows Zhangsan NEG work.out
#suoyi Zhangsan you bu jianshen.
therefore Zhangsan YOU NEG work.out
Intended: 'What are you talking about? Everyone knows that Zhangsan doesn't work out, therefore he doesn't.'

Second, conventional implicatures must project globally, even in the consequent of conditionals, whereas presuppositions allow a local interpretation (Karttunen 1973; Potts 2004). If (67B) contains *you*, it is odd to continue it with '... he will have diabetes.' But if (67B) doesn't contain *you*, then this continuation is fine.

- (67) A: 'Zhangsan has a chance of having diabetes in the future.'
B: ruguoZhangsan meitian jianshen,
if Zhangsan every.day work.out
name ta #you bu hui de tangniaobing.
then he YOU NEG will get diabetes
dan kexi ta conglai bu jianshen, suoyi ta hui de.
but it's.a.shamehe ever NEG work.out, therefore he will get
'If Zhangsan works out every day, then he won't have diabetes. But it's a shame that he never works out, so he will have diabetes.'

The reason why the sentence containing *you* cannot have this continuation is because *you*'s meaning projects out of the conditional. In (67B) *you* requires its prejacent (i.e. Zhangsan won't have diabetes) to be in the Common Ground, which contradicts the continuation. If *you* could require the entire conditional (i.e. if Zhangsan works out every day, then he won't have diabetes)

to be in the Common Ground instead, then it should be compatible with the continuation, contrary to fact.

It is difficult to test whether *bing*'s meaning can be interpreted locally in conditionals because it is difficult to tease apart its local and global interpretations. The local interpretation would be: *if p, then bing not q* requires there to be a salient proposition *if p, then q*; the global interpretation would be: *if p, then bing not q* requires there to be a salient proposition *q*. Satisfying the local interpretation entails satisfying the global interpretation because whenever there is a salient proposition *if p, then q*, then *q* is also salient.

In sum, this section has argued that *bing* presupposes that its prejacent contrasts with a salient proposition, while *you* has the conventional implicature that its prejacent was already in the Common Ground.

It is worth mentioning that besides correction, *bing* and *you* have other uses in the language, which are deeply related to correction under the current view. Both *bing* and *you* can be used as coordinators of predicates 'and', and in addition *you* can mean 'again', etc. (see footnote 2). In these other uses, *bing* and *you* are not corrective and can be used out of the blue:

(68)a. Zhangsan tong bing kuaile-zhe.
Zhangsan painful BING happy-ASP
'Zhangsan is both in pain and happy.'

(69)a. Zhangsan you changge you tiaowu.
Zhangsan YOU sing YOU dance
'Zhangsan danced and sang.'

b. Zhangsan you changge le
Zhangsan YOU sing PERF
'Zhangsan sang again.'

It may not be accidental that *bing* is a coordinator. Section 4.1 has highlighted the similarities between *bing* and English counterexpectational *but*, which is a coordinator. I thus speculate that *bing* was originally 'and', and over time developed the counterexpectational meaning. I will continue to explore the parallel between *bing* and the coordinator *but* in section 7, where I essentially argue that many languages (including English) have an operator equivalent to *bing* that introduces the first conjunct of corrective 'but' sentences. Under this view, we can consider this *bing*-like operator to parallel corrective *but* in that they introduce the first and second conjuncts respectively. Thus, the *bing*-like operator is a coordinator. As for *you*, as a corrective marker it reminds the hearer of information already in the Common Ground, and thus can be viewed as meta-conversational 'again'.

5. Previous literature on *bing* and *you*

Bing and *you* have been discussed extensively in the Chinese literature, which largely makes empirical generalizations about their use rather than formal semantic and pragmatic proposals. These generalizations are not completely accurate and fail to capture the full range of data. This subsection discusses five main generalizations, and raises counterexamples to each one of them. Then I will discuss Yuan & Liu's (2025) formal semantic-pragmatic proposal and compare it with the current one.

First, Wang (2001), Yin (2011), Cheng (2016) and Wen & Li (2019) argued that *bing* is used for counterexpectation while *you* is used to express the reason. But *bing* can be embedded in ‘because’-clauses (37), and *you* can be embedded in counterexpectational ‘but’-clauses:

- (70) A: ‘I’m hungry.’
 B: danshi fandian you mei you kai
 but restaurant YOU NEG have open
 ‘But the restaurants aren’t open.’

Second, Shi (1990) and Li (2014) argued that *bing* marks a proposition that the speaker expects, while *you* marks a proposition that the hearer does not expect. But *bing* can mark a proposition that is surprising to the speaker (highlighted by *jingran* ‘surprisingly’ in (71)), and *you* can mark a proposition that the hearer expects as in (72), where the mother (speaker B) certainly knows that her child has hands.

- (71) Zhangsan meitian qu jianshenfang,
 Zhangsan every.day go gym
 danshi ta jingran bing bu jianshen.
 but he surprisingly BING NEG work.out
 ‘Zhangsan goes to the gym every day, but surprisingly, he doesn’t work out.’

- (72) A: ‘Mom, I’m hungry.’
 B: ni you bu shi mei you shou, ziji qu cheng fan!
 you YOU NEG COP NEG have hand self go get rice
 ‘It’s not like you don’t have hands. Go get rice yourself!’

Third, Chen (1987, 2019), Peng (1999), Shi (2005), Wen (2009), Wu (1999) and Yang (2008) argued that *bing* expresses an objective fact, while *you* expresses a subjective opinion and correction. But *bing* can express a subjective opinion (73), and *you* can express an objective fact (70).

- (73) meigeren dou xiang chengwei mayun, danshi wo bing bu xianmu ta.
 everyone all want become Jack.Ma but I BING NEG envy him
 ‘Everyone wants to become Jack Ma, but I don’t envy him.’

Fourth, Chen (2021) and Nie (2021) argued that *bing* denies a fact, while *you* denies how things should be. But *bing* can express how things should be (74), while *you* can deny a fact (70).

- (74) ni bing bu yinggai zheme zuo.
 you BING NEG should so do
 ‘You shouldn’t do this.’

The fifth generalization is an influential view in the Chinese literature: *bing* marks direct denial, while *you* marks indirect denial by denying the precondition of a proposition or speech act (Ma 2001; Liu 2010; Zhang and Liu 2010; Zhang and Yan 2015). But *bing* can mark indirect denial (46), (47) and (51), and *you* can mark direct denial (2). While this Chinese literature would

consider sentences like (2) with *you* to be odd, my consultants and I think they are fine with surprised intonation, putting strong prosodic focus on the subject:

- (75) A: ‘Zhangsan works out.’
 B: Zhangsan_F you bu jianshen.
 Zhangsan YOU NEG work.out
 ‘Zhangsan doesn’t work out. (What are you thinking?)’

This special prosody expresses speaker B’s surprise at speaker A’s statement. I think *you* in direct denial requires this prosody precisely because of *you*’s meaning: speaker B believes that it was already in the CG that Zhangsan doesn’t work out, and thus is shocked to hear direct evidence against their assumption (i.e. speaker A says Zhangsan works out). This special prosody is possible but not required when *you* is used for indirect denial because the hearer has not directly contradicted the speaker’s belief about the CG as they do in direct denial. For example, in the indirect denial in (70), based on speaker A’s utterance, speaker B can but does not need to infer that speaker A believes that the restaurants are open. There are many ways to satisfy speaker A’s hunger other than going to a restaurant, such as cooking at home or getting snacks at a grocery store. Thus, speaker B does not need to use the surprised intonation because they have not seen direct evidence that speaker A believes the restaurants are open.

Yuan & Liu (2025) built on the fifth empirical generalization (i.e. *bing* is for direct denial, while *you* is for indirect denial), and proposed an analysis that is similar to mine but with key differences: they claimed that *bing not p* presupposes that *p* is old information, which they considered to be equivalent to presupposing that someone believes that *p* is more than 50% likely, while *you not p* has the conventional implicature that $\neg p$ is already in the CG, and presupposes that the speaker thinks that *p* is the sufficient condition of some existing act (speech act or physical act). There are two key differences between their analysis and mine. First, they argued that *bing not p* presupposes that someone believes *p* is likely, whereas I do not think people’s belief is relevant here. The following examples show that *bing not p* does not require anyone to believe *p*:

- (76) suoyouren dou zhidao zhangsan bu jianshen,
 everyone all know Zhangsan NEG work.out
 shishiye shi ta bing bu jianshen.
 fact also COP he BING NEG work.out
 ‘Everyone knows that Zhangsan doesn’t work out, and the fact is also that he doesn’t work out.’
- (77) A: meiren renwei Zhangsan jianshen.
 no.one think Zhangsan work.out
 ‘No one thinks that Zhangsan works out.’
 B: dui, ta bing bu jianshen.
 right he BING NEG work.out
 ‘Correct, he doesn’t work out.’

The second difference between their analysis and mine is that they proposed an additional presupposition for *you–p* should be the sufficient condition of some existing act. I do not think this presupposition is necessary because it follows from its conventional implicature. Because

you implies the speaker's belief about the CG, by the maxim of quantity, the speaker would not repeat what they believe is old information (i.e. what is in the CG) unless they see evidence against their belief. In other words, the use of *you* requires the speaker to see evidence that contradicts their belief about the CG (i.e. evidence that implies that $\neg p$ is not in the CG). This evidence can be the hearer's act for which p is the sufficient condition because based on this act, the speaker infers that the hearer believes p , which contradicts the speaker's belief that $\neg p$ is in the CG. This evidence can also be p itself, in which case the speaker says *you not p* in direct denial of p . One of Yuan & Liu's arguments for their proposed presupposition of *you* is that *you* cannot be used in direct denial, but (75) showed that it can, as long as there is surprised prosody.

This section has shown counterexamples to five empirical generalizations about *bing* and *you* in the Chinese literature. I have argued that the current proposal is sufficient for accounting for the distribution of *bing* and *you*, and that we do not need the presupposition of someone's belief for *bing* or the presupposition of precondition for *you* as Yuan & Liu's (2025) claimed.

6. Comparison with Italian *mica* and Hindi-Urdu *thorī*:

Besides *bing* and *you*, there is a small but growing literature on corrective markers in other languages, such as *mica* in Italian and *thorī*: in Hindi-Urdu. (7)–(8) demonstrated this requirement of *mica* requires a salient proposition to correct, and the following examples demonstrate this requirement of *thorī*:

(78) Context: You ask me to tell you something about my friend Mayank, who you don't know anything about. I say:

Mayank	roz	gym	{nahĩ:/#thorī:}	ja:-ta:	hai
Mayank	every.day	gym	NEG/THORI	go-IPFV.MSG	be.PRS.3SG
'Mayank doesn't go to the gym every day.'					(Bhatt & Homer 2022:2)

(79) A: 'Mayank goes to the gym every day.'

B:	Mayank	roz	gym	{nahĩ:/thorī:}	ja:-ta:	hai
	Mayank	every.day	gym	NEG/THORI	go-IPFV.MSG	be.PRS.3SG
	'Mayank doesn't go to the gym every day.'					(Bhatt & Homer 2022:2)

Mica and *thorī*: are negative in nature: they can co-occur with negation, but when they don't, the sentence has the same meaning as a sentence with negation. Frana & Rawlins (2019) (F&R) analyzed the sentences without negation as incorporation of negation into *mica*.

There are two types of proposals for *mica* and *thorī*:, which bear similarities to my proposals for *bing* and *you* respectively. As an example of the first type, Cinque (1976) argued that *mica* requires a prior claim or a salient expectation to deny. Although he did not formalize what it means to deny a salient proposition, we could formalize it in terms of contrast like I did in section 4.1, and consider this analysis to be identical to my proposal for *bing*. As an example of the other type of proposals, F&R argued that *mica* presupposes that the speaker is sure that p should not be added to the CG, thus making *mica* a CG manager, in the same vein as my proposal for *you*. Bhatt & Homer (2022) built on F&R's analysis, and added focus sensitivity to it: *thorī*: presupposes that the speaker is sure that p should not be added to the CG, but instead an alternative p' should be.

This section compares the behaviors of *bing* and *you* with those of *mica* and *thori*:, with a focus on two types of data: (a) key data that teased apart the contrast-marker analysis and the CG-manager analysis of *mica*, which when applied to *bing*, suggest that *bing* is a contrast marker (subsection 6.1); and (b) key data that motivated the CG-manager analysis of *mica*, which when applied to *bing* and *you*, suggest that *bing* is not a CG-manager but a contrast marker, and *you* is a CG-manager with different meanings from *mica* and *thori*: (subsection 6.2).

6.1. Key data that teased apart the contrast-marker analysis and the CG-manager analysis of *mica*: The ‘elevator’ dialogues

F&R raised the following evidence against Cinque’s contrast-marker analysis of *mica*: assuming that residential buildings normally have elevators but not escalators, in response to an IKEA delivery person A’s question, it is rude to put *mica* in the elevator-answer (80a) but okay to put it in the escalator-answer (80b).

- (80) a. A: ‘Does your building have an elevator?’
 B: No, il mio palazzo non ha (#mica) un ascensore.
 ‘No, my building does not have (#mica) an elevator.’
- b. A: ‘Does your building have an escalator?’
 B: No, il mio palazzo non ha (mica) una scalamobile.
 ‘No, my building does not have (mica) an escalator.’ (Frana & Rawlins 2019:47)

If *mica* presupposes its prejacent is a salient expectation as Cinque claimed, the delivery person’s question should license this presupposition in both answers, contrary to fact. This fact can be captured by F&R’s proposal that *mica* presupposes that the speaker is sure that *mica*’s prejacent should not be added to the CG: only in (80b) is the answerer justified to assume that the proposition that their building has an escalator should not be in the CG because this is not a common proposition to assume. The proposition that buildings have elevators is more commonly assumed, and thus the answerer in (80a) is not justified in assuming that this proposition should not be in the CG.

Bing is possible in both answers, precisely what Cinque would predict for Italian *mica*:

- (81) a. A: ‘Does your building have an elevator?’
 B: wode dalou {bing/#you} mei you zhiti.
 my building BING/YOU NEG have elevator
 ‘My building does not have an elevator.’
- b. A: ‘Does your building have an escalator?’
 B: wode dalou {bing/#you} mei you futi.
 my building BING/YOU NEG have escalator
 ‘My building does not have an escalator.’

This supports the current analysis that *bing* requires its prejacent to contrast with a salient proposition. This salient proposition is *speaker B’s building has an elevator / escalator*, which was made salient by the delivery person’s question.

In contrast, *you* is odd in both answers, suggesting that its analysis should differ from F&R's analyses for *mica*. I argue that *you* is odd in both answers for the same reason that (59) is rude: it is rude to assume the delivery person should know whether speaker B's building has an elevator / escalator. Even if buildings normally don't have escalators, it is rude to assume the delivery person knows this fact about speaker B's building in particular.

According to F&R's analysis, *mica* talks about what the speaker believes *was not* in the CG, whereas in my analysis, *you* talks about what the speaker believes *was already* in the CG. The user of *you* holds a much stronger belief than the user of *mica* because a proposition not yet in the CG may be normally false, or hasn't been fully discussed by the interlocutors, but may eventually be added to the CG after their discussion of it, therefore the user of *mica* is open to eventually adding that proposition to the CG. But a proposition already taken to be in the CG cannot be easily removed, unless the user of *you* is willing to withdraw their belief about the CG and re-negotiate with the hearer about the proposition. The fact that *you*'s requirement is much stronger than *mica*'s has the effect that *you* has an impatient and even rude connotation that *mica* does not have.

6.2. Key data motivating the CG-manager analysis of *mica*: Negative polar questions

The key data that motivated F&R's analysis was negative polar questions (NPQs), which normally require the speaker to have a prior bias for *p*. But when *mica* appears in NPQs, this bias is reversed to $\neg p$. The context in (82) requires the speaker's prior bias for *p*, and *mica* is not possible; the context in (83) requires prior bias for $\neg p$, and *mica* is required.

(82) Context (Speaker expected *p*, but sees evidence for $\neg p$): Clara invites Miles for drinks and tells him to come after dinner. When he gets there, Miles asks if she has any food. Clara asks:

- a. Non hai già mangiato? (NPQ)
NEG have.2sg already eaten?
'Didn't you eat already?'
- b. #Non hai mica già mangiato? (*mica*-NPQ)
NEG have.2sg mica already eaten? (Frana & Rawlins 2019:18)

(83) Context (Speaker expected $\neg p$, but sees evidence for *p*): Clara invites Miles for dinner and makes clear to him that she will prepare her best dishes. When he gets there, Miles barely touches any food. Clara asks:

- a. #Non hai già mangiato? (NPQ)
- b. Non hai mica già mangiato? (*mica*-NPQ)
(Frana & Rawlins 2019:18)

Based on this contrast between (82) and (83), F&R proposed that *mica* presupposes that prior to their question, the speaker believed *p* should not be added to the CG. This is satisfied in (83), where the speaker believed $\neg p$ to be in the CG, but not satisfied in (82).

I claim that *you* and *bing* have different meanings from *mica*, and thus predict NPQs containing *you* and *bing* to behave differently from those containing *mica*. This prediction is borne out, as the following subsections evaluate *you* and *bing* in NPQs respectively.

6.2.1. *You* in NPQs

You cannot be embedded in NPQs at all:

- (84) #Zhangsan you bu jianshen ma?
Zhangsan YOU NEG work.out Q

I argue that this is because unlike *mica*, whose user thinks *p* should not be added to the CG but is open to negotiation, the user of *you* believes that this negotiation has already happened, and that $\neg p$ is already in the CG, and therefore cannot retract that negotiation by asking an NPQ.

6.2.2. *Bing* in NPQs

Bing can appear in NPQs, but before evaluating *bing* in NPQs, I first discuss the meaning of an NPQ without *bing*, and then show that an NPQ with *bing* has the same meaning, plus the additional presupposition by *bing*, which is expected under the current analysis. Polar questions in general vary in three aspects: (a) what sort of prior bias was held by the speaker before asking the question; (b) what sort of contextual evidence was presented to the speaker before asking the question; and (c) what proposition the question checks or double-checks.

My proposal is that *bing* presupposes that its prejacent contrasts with some salient proposition in the context. This should not affect any of these three aspects of polar questions. Thus, I predict that NPQs containing *bing* should behave the same as those without *bing* in these three aspects, but in addition, NPQs containing *bing* require its prejacent to contrast with some salient proposition. I will now show that this prediction is borne out. First, let us consider a context that does not involve any prior bias by the speaker (85), adapted from F&R.

(85) Context: S interviews A on TV about Wu Guanzhong, a Chinese painter.

A: ‘There were some times in Mr. Wu’s career when he only painted landscapes, but not portraits.’

S: ‘Please tell us more about those gaps. For example...’

qishi niandai ta (#bing) mei you hua renhe renwuhua ma?

seventy times he BING NEG₂ have paint any portrait Q

rugomei you, na shi weishenme ne?

if NEG have then COP why PRT

‘did he not paint any portrait in the 70s? And, if he didn’t, why not?’

(context without speaker’s prior bias, adapted from Frana & Rawlins 2019:23)

As is demonstrated by the licensing of the NPI here, the negation in (85) is Neg₂, and thus this NPQ does not require the speaker’s prior bias. Like its English translation *Did he not paint any portrait in the 70s?*, this Mandarin NPQ checks the truth and falsity of a negative proposition. *Bing* is not licensed in this context because *bing*’s presupposition is not satisfied: *He painted portrait in the 70s* is not a salient proposition. We can fix this by changing the context slightly:

(86) Context: S interviews A on TV about Wu Guanzhong, a Chinese painter. S knows about his portrait gaps in the 70s, and says ‘Because Mr. Wu is a versatile painter, some may expect him to paint landscapes and portraits throughout his career...’

dan qishi qishi niandai ta (bing) mei you hua renhe renwuhua ma
 but actually seventy times he BING NEG2 have write any portrait Q
 ‘but actually, did he not paint any portrait in the 70s?’

S’s prior utterance makes salient the proposition that contrasts with *bing*’s prejacent, and thus licenses *bing* in (86). To summarize, I have shown that *bing* and *you* behave differently from *mica*: *bing* behaves like what Cinque predicts *mica* would behave, while *you* is a CG manager, like F&R argued for *mica*, but *you* makes a different contribution to the discourse from *mica*.

Besides the differences in their detailed semantics, there is a key syntactic difference between *bing* and *you* on the one hand and *mica* on the other: F&R put *mica* in the left-peripheral CP because it is a meta-conversational operator. In their analysis, the negation associated with *mica* is also interpreted high because rather than asserting a negative proposition, the speaker *refuses* to add a proposition to the CG. In section 2 I have shown that *bing*, *you* and negation are located lower than *mica*, between C and T, and section 3 has argued that negation is interpreted in-situ—*bing* and *you* make meta-conversational moves based on a negative proposition. I therefore suggest that the illocutionary force associated with *bing* and *you* is actually introduced by covert operators in Force in the left-peripheral CP that agree with *bing* and *you*, and *bing* and *you* are semantically inert.

7. Corrective markers in the languages that don’t seem to have one

I have focused on the corrective markers in Mandarin, Italian and Hindi-Urdu so far. Many languages such as English don’t seem to have a dedicated word for correction, but I want to argue that they actually do—this corrective marker is equivalent to Mandarin *bing*. It occurs in the first conjunct of corrective ‘but’ sentences, but is silent. Due to this silent marker, the first conjunct of corrective ‘but’ sentences shares many properties with Mandarin *bing*-sentences. Consider English corrective *but* sentences as an example. They cannot be uttered out of the blue, but require there to be a proposition to correct:

(87) Context: I tell you about my friend Zhangsan, who you don’t know anything about. I say:
 #Zhangsan doesn’t work out, but goes on long walks.

(88) A: Zhangsan works out.
 B: He doesn’t work out, but goes on long walks.

As mentioned in section 4.1, the first conjunct of corrective *but* requires negation. This parallels *bing* and *you*’s requirement of negation. When correcting a negative proposition, the first conjunct of corrective *but* still requires negation, leading to double negation:

(89) A: Zhangsan doesn’t work out.
 B1: *He works out, but goes on long walks.
 B2: He doesn’t not work out, but lifts weights.

While English corrective *but* has the same morphology as the other uses of *but*, some languages have a dedicated lexical item for corrective ‘but’ (e.g. *sondern* in German, *sino* in Spanish, *ela* in Hebrew and *amma* in Persian). These corrective ‘but’s all requires negation in their first conjunct and absence of negation in the second. Consider the following German sentence as an example:

- (90) Das ist nicht bewusst, sondern ganz automatisch.
 this is NEG conscious but completely automatic
 ‘This is not conscious, but completely automatic.’ (Anscombe & Ducrot 1977:2)

Due to this morphological difference between corrective ‘but’ and the other uses of ‘but’ in these languages, along with other differences in meaning and distribution, (Anscombe and Ducrot 1977) proposed that English and French also have dedicated corrective ‘but’, which are *but* and *mais* respectively, despite their morphological identity to the other ‘but’s in the language.

All the existing research that I know of has attributed the corrective meaning of corrective ‘but’ sentences entirely to ‘but’. For example, Anscombe and Ducrot (1977) focused on the contrast between corrective ‘but’ and non-corrective ‘but’, and proposed that corrective ‘but’ requires a previous assertion of *p* in the discourse. This analysis is very similar to mine of *bing*, except that their suggestion is stronger because they did not consider the cases involving implication. (Kasimir 2008) claimed that the negation in German ‘not...but...’ sentences is ordinary truth-functional negation, and *sondern* ‘but’ encodes all the corrective meaning: it requires its conjuncts to be unmarked answers to some question. This analysis is similar to mine in the adoption of the notion of question-answerhood.

I will discuss many similarities between *bing* and corrective ‘but’ sentences in subsection 7.1.1, based on which I extend my analysis of *bing* to corrective ‘but’ sentences. Contrary to the literature, I don’t put the corrective meaning of corrective ‘but’ sentences on ‘but’, but on a silent marker in the first conjunct. In languages that don’t seem to have a dedicated corrective marker but have corrective ‘but’ sentences, there is a silent Op(erator) equivalent to Mandarin *bing* in the first conjunct of these corrective sentences, and this silent Op requires negation, just like *bing* does:

- (91) a. Zhangsan does Op not work out, but goes on long walks.
 b. Das ist Op nicht bewusst, sondern ganz automatisch.

It is worth clarifying that I do not say corrective *but* contributes no meaning at all. It does by marking contrast between its two conjuncts, whereas the silent Op marks contrast between the first conjunct and a salient proposition. We can adopt Toosarvandani’s (2014) analysis of corrective *but*, where corrective *but* presupposes that its conjuncts entail (rather than just imply) the answers to a QUD that contrast in polarity.

My analysis thus presents a unifying view of *bing*, Op and *but*: they all mark contrast between two propositions. *Bing* and Op contrast their prejacent with a salient proposition, whereas counterexpectational *but* and corrective *but* contrast their two conjuncts. Following Toosarvandani (2014), the strength of implication of these contrast markers is different: *bing*, Op and counterexpectational *but* only require contrast in terms of implication (i.e. entailment in the most normal worlds), whereas corrective *but* requires contrast in terms of entailment in all worlds.

Subsection 7.1.1 will provide five arguments for my analysis by showing that the first conjunct of corrective ‘but’ behaves the same as Mandarin *bing*-sentences, using English corrective *but* sentences as examples. Then I will discuss a difference between Mandarin *bing* and the silent corrective marker Op: Op requires ‘but’, whereas Mandarin *bing* doesn’t. Subsection 7.1.2 will then claim that it may be universal that all corrective markers require negation, reject three semantic hypotheses for why this is, and instead suggest a syntactic analysis, where corrective markers syntactically select for negation.

7.1.1. Corrective negation behaves like *bing*

The beginning of section 7 has presented two properties that *bing*-sentences and the first conjunct of corrective *but* have in common: neither of them can be uttered out of the blue, but requires a proposition to correct; and they both require negation. Besides these two properties, there are five other aspects in which they behave the same. Before presenting them, it is worth clarifying that as I will argue at the end of this subsection, English corrective negation requires a *but*-clause, and thus all the English examples involve a *but*-clause. But what behaves like *bing*-sentences is the first conjunct of corrective *but* sentences because it requires negation. The first aspect in which the first conjunct of corrective *but* behaves like *bing* (and *you*) is that it can correct not at-issue content and even non-linguistic statements:

- (92) a. A: How is your dog?
B: I don’t have a dog, but a cat.

- (93) Context: A puppy approaches Zhangsan, who looks scared. I tell Zhangsan:
Don’t be afraid. It won’t bite you, but only greet you.

Second, like *bing* but not *you*, corrective *but* sentences can correct the expectation created in the same sentence:

- (94) Context: You ask me to tell you something about my friend Zhangsan, who you don’t know anything about. I say:
Zhangsan goes to the gym every day, but he doesn’t work out, but observes people.

Third, like *bing* but not *you*, corrective *but* sentences can be embedded in polar questions and finite clauses:

- (95) a. Does Zhangsan not work out, but go on long walks?
b. A: I hope Zhangsan will come to the party.
B: If Zhangsan won’t come to the party but stay at home, then there’s no point in hosting it.
c. Because Zhangsan doesn’t work out but stay at home, he risks having diabetes.
d. Lisi {said/thought/was surprised/realised} that Zhangsan doesn’t work out but stay at home.

Fourth, like *bing* but not *you*, corrective *but* sentences does not have the impatience implication, and can appear in both the elevator and escalator answers discussed in section 6:

- (96) a. A: Does your building have an elevator?
 B: My building does not have an elevator, but an escalator.
- b. A: Does your building have an escalator?
 B: My building does not have an escalator, but an elevator.

Fifth, like *bing*, corrective *but* sentences can be used to deny an implication:

- (97) A: I'm hungry.
 B: But the restaurants aren't open now, but this time tomorrow.

This suggests that even though there is no overt word equivalent to *bing* in the first conjunct of corrective *but* sentences, there is a silent Op equivalent to *bing* that presupposes that its prejacent should contrast with a salient proposition (e.g. *My building does Op not have an elevator, but an escalator*).

All the Mandarin examples in this paper so far do not involve a 'but'-clause, even though they can, suggesting that *bing* does not require 'but':

- (98) A: 'Zhangsan works out.'
 B: Zhangsan bing bu jianshen, er shi xihuan sanbu.
 Zhangsan BING NEG work.out but COP like walk
 'Zhangsan doesn't work out, but likes to take walks.'

In contrast, the silent Op in English requires *but*. English correction can involve two types of negation—constituent negation and sentence negation. Constituent negation always requires *but*:

- (99) a. *John eats not spinach.
 b. John eats not spinach but chard.

Sentence negation doesn't require *but*, but without *but*, we lose the corrective meaning and the associated properties. For example, it doesn't require a proposition to correct:

- (100) Context: I tell you about my friend Zhangsan, who you don't know anything about. I say:
 Zhangsan is religious. He doesn't eat meat.

Conversely, without the *but*-phrase, negation is not required, either:

- (101) A: Zhangsan doesn't work out.
 B: Zhangsan does work out.

This can be accounted for by the current analysis, where the *but*-phrase is required by the silent corrective marker Op, which also requires negation. Therefore, I assume that Op in English always requires *but*, whereas Mandarin *bing* doesn't. Except for this difference, Op has the same distribution as *bing*.

7.1.2. Corrective markers select for negation

We saw that *bing* and *you* require negation. English corrective *Op* also requires negation, as does the first conjunct of corrective ‘but’ sentences across all the languages that I know of (e.g. French, German, Hebrew, Persian and Spanish, which include unrelated languages). Thus, I will assume that it is a universal property of corrective markers that they require negation. This subsection will present examples of corrective ‘but’ sentences in English, French and German showing that what is required is syntactic negation, not incorporated negation. Then I will discuss three possible semantic hypotheses of why correction requires negation, and show that none of them works. I will end with the suggestion that corrective markers syntactically select for negation.

It has been observed that corrective ‘but’ in languages like English, German and French requires negation in the first conjunct, and this negation cannot be incorporated negation (Anscombe and Ducrot 1977); Horn 2001; (Kasimir 2008); Toosarvandani 2014):

(102) Space travel is {not possible/#impossible}, but a dream. (Toosarvandani 2014:43)

(103) Einstein’s approach was {not religious/#irreligious}, but rational. (Horn 2001:392)

(104) Er ist {nicht freundlich/#unfreundlich}, sondern ziemlich unhöflich.
 Heis NEG friendly unfriendly but rather impolite
 ‘He is {not friendly/#unfriendly} but rather impolite.’ (Kasimir 2008:136)

(105) Il {n’est pas intelligent/#est inintelligent}, mais seulement bûcheur.
 He is.NEG NEG intelligent is unintelligent but only hard.working
 ‘He is {not intelligent/#unintelligent} but only hard working.’
 (Anscombe and Ducrot 1977, 35)

In parallel, *bing* and *you* require negation, but not incorporated negation. Other than *bu* and *mei*, Mandarin has negative auxiliaries and verbs, such as *wei*, the negated perfective and *wu*, ‘not have / exist’. These markers are sentence negation syntactically, and can satisfy *bing*’s negative requirement:

- (106) a. wo bing wei fan cuo.
 I BING PERF.NEG make mistake
 ‘I didn’t make any mistake.’
 b. wo bing wu eyi.
 I BING have.NEG bad.intentions
 ‘I don’t have bad intentions.’

The negated perfective *wei* and *wu* ‘not have / exist’ can be incorporated into negative adverbs such as *weibi* ‘not necessarily’, *weiceng* ‘not ever’ and *weichang* ‘not ever’ and the negated modal *wuxu* ‘need not’. *Bing* cannot appear with these derived negative words:

- (107) a. *wo bing weibi hui shu.
 I BING not.necessarily will lose
 Intended: 'I won't necessarily lose.'
- b. *wo bing {weiceng/weichang} fan cuo.
 I BING not.ever make mistake
 Intended: 'I didn't ever make any mistake.'
- c. *ni bing wuxu duo yan.
 you BING need.NEG a.lot say
 Intended: 'You don't need to say much.'

Toosarvandani (2014) analyzed the English fact this way: the first conjunct of corrective *but* is the answer to a negative question under discussion (i.e. *What was Einstein's approach not?* and *What is space travel not?*). Negative questions in general cannot be answered by incorporated negation:

- (108) A: Space travel might be possible, or it might be a dream. What is space travel not?
 B1: It's not possible.
 B2: #It's impossible.

Kasimir (2008) hinted at a similar answer-based analysis. Such an analysis does not work for Mandarin because the sentences with incorporated negation in (107) can be the answers to the relevant negative questions:

- (109) a. Q: Ni bu yiding hui zenmeyang?
 you NEG necessarily will what
 'What won't you necessarily do?'
 A: wo weibi hui shu.
 I not.necessarily will lose
 'I won't necessarily lose.'
- b. Q: Ni mei you zuo-guo shenme?
 you NEG have do-PRF what
 'What haven't you done?'
 A: wo {weiceng/weichang} fan cuo.
 I not.ever make mistake
 'I haven't ever made any mistake.'
- c. Q: Wo bu xuyao zuo shenme?
 I NEG need do what
 'What will I no longer need to do?'
 A: ni wuxu duo yan.
 you need.NEG a.lot say
 'You don't need to say much.'

Another type of words that require negative environments is NPIs, and thus another possible explanation for why corrective markers require negation is that they are NPIs. But corrective markers do not have the same distribution as NPIs. Incorporated negation in English and Mandarin can license NPIs, in contrast to its inability to license corrective markers:

(110) It is impossible to imagine any other scenario.

(111) a. wo weibi hui chi renhe kui.
 I not.necessarily will suffer any loss
 ‘I won’t necessarily suffer any loss.’

 b. wo {weiceng/weichang} fan renhe cuo.
 I not.ever make any mistake
 ‘I didn’t ever make any mistake.’

 c. ni wuxu duo shuo renhe hua.
 you need.NEG a.lot say any word
 ‘You don’t need to say much.’

My analyses of *bing* and *you* in this paper do not predict that they require negation. *You* only implies that its prejacent was already in the Common Ground, but nothing in my analysis requires this proposition already in the Common Ground to involve syntactic negation. *Bing* only requires its prejacent to contrast with a salient proposition, and a sentence without syntactic negation can perfectly fulfill this requirement. For example, ‘Zhangsan works out’ contrasts with the salient proposition ‘Zhangsan doesn’t work out’, as in (3). One might hypothesize that contrast somehow inherently requires syntactic negation, but recall from section 4.1 Winter and Rimon’s (1994) analysis of contrast, which is really about contrasted polarities in the “proposition under debate” (i.e. r and $\neg r$ in Figure 2). Therefore, this hypothesis would only require syntactic negation in that proposition under debate, but not in *bing*’s prejacent, which only needs to imply that proposition (i.e. entail $\neg r$ in the most normal worlds). For example, let us consider (51) again, where (51B) implies that speaker A will not get to eat:

(51) A: ‘I’m hungry.’
 B: (danshi) fandian bing mei you kai
 but restaurant BING NEG have open
 ‘The restaurants aren’t open.’

Replacing *bing*’s prejacent with the positive proposition ‘the restaurants are closed’ leads to ungrammaticality (**danshi fandian bing guanzhe a* ‘But the restaurants bing are closed’), but this sentence like (51B) implies a proposition with syntactic negation–speaker A will not get to eat.

After arguing that there is a *bing*-like silent corrective marker in the first conjunct of corrective ‘but’ sentences, I have shown that it may be a universal property of corrective markers that they require syntactic negation, discussed three different hypotheses for why, and showed that none of them works. Toosarvandani (2014) argued that answers to negative questions under discussion must involve syntactic negation rather than incorporated negation, but this is not the case in Mandarin Chinese. Corrective markers are not NPIs because they

have different distributions. My analyses of *bing* and *you* do not predict that they require negation, either, even if we hypothesize in addition that contrast requires syntactic negation. Therefore, I suggest for now that corrective markers syntactically select for negation, and leave a possible alternative explanation to future research.

8. Conclusion

I have argued that among the three negations on the clausal spine, *bing* and *you* require the middle one, Neg₂, putting them between C and T. Different meanings and distributions of *bing* and *you* led me to propose that the illocutionary force associated with *bing* presupposes that its prejacent contrasts with a salient proposition, while the illocutionary force associated with *you* implies that the speaker believes that its prejacent was already in the CG. Then I enriched the typology of corrective markers with two broad comparisons. I compared *you* to *mica* and *thori:*, which are all CG managers, but *you* encodes a stronger belief about the CG, leading to its impatient and sometimes rude connotation. I compared *bing*-sentences with counterexpectational *but* and corrective *but* sentences, and argued that there is a silent Operator equivalent to *bing* in the first conjunct of corrective *but* sentences. This leads to unification of *bing*, this silent Op and ‘but’: they all mark contrast, but differ in which propositions they contrast: the former two contrast their prejacent with a salient proposition, while the latter contrast their conjuncts. The following table summarizes the meanings of these corrective markers.

Type	Corrective marker	Meaning
Contrast marker	<i>Bing</i>	Presupposition: its prejacent and a salient proposition should imply yes- and no- answers to the QUD.
	Counterexpectational <i>but</i>	Presupposition: its prejacent and the first conjunct of <i>but</i> should imply yes- and no- answers to the QUD.
	Silent Op in the first conjunct of corrective ‘but’ sentences	Presupposition: its prejacent and a salient proposition should imply yes- and no- answers to the QUD.
	Corrective ‘but’	Presupposition: its prejacent and the first conjunct of ‘but’ should entail yes- and no- answers to the QUD.
Common Ground manager and/or Evidential marker	<i>You</i>	Conventional implicature: the speaker believes that its prejacent was already in the Common Ground.
	<i>Mica</i>	Presupposition: the speaker is sure that its prejacent should not have been added to the CG.
	<i>Thori:</i>	Presupposition: the speaker is sure that its prejacent should not have been added to the CG, but instead an alternative proposition should have been.

Table 3: Typology of corrective markers.

This paper began with the observation that *bing* and *you* require a proposition to correct (1)–(2), making them corrective markers like *mica* and *thorì*. But I have subsequently shown that corrective markers, as are exemplified by *bing* and *you*, fall into two categories—contrast markers and Common Ground managers. These two types all require a proposition to correct, but for different reasons: this requirement is part of contrast markers such as *bing*’s presupposition because it requires contrast to a salient proposition. In contrast, the *you*-sentence reiterates a proposition that the speaker thinks was already in the CG. Therefore, unless the hearer’s act provides evidence that they believe in the contrary proposition, *you* is not licensed.

References

- Anscombe, Jean-Claude, and Oswald Ducrot. 1977. Deux mais en français? *Lingua* 43: 23–40.
- Bhatt, Rajesh, and Vincent Homer. 2022. Negation of disagreement in Hindi-Urdu presented at the Harvard Linguistics Circle, Harvard University.
- Biq, Yung-O. 1988. From objectivity to subjectivity: The text-building function of *you* in Chinese. *Studies in Language* 12: 99–122.
- Chen, Yu. 2019. Xiandai hanyu keguanxing fanchou [The objective range of modern Chinese]. *Hanyu Xuexi (Chinese Language Learning)*: 9.
- Chen, Yu. 2021. Differentiation of negation from the perspective of state of affairs: A case study on the construction competition between “bingbu x” and “youbu x.” *Journal of Foreign Languages* 44: 2–10.
- Chen, Yueming. 1987. “You” de yizhong yufa yi yi xin jie [New interpretation of a kind of semantics of ‘you’]. *Yuyan Jiaoxue yu Yanjiu (Language Teaching and Linguistic Studies)*: 5.
- Cheng, Yaheng. 2016. Yuanyinxing “you+Neg+Xp” goushi de huihua gongneng (The conversational functions of reason-type “you+Neg+Xp” structure). *Hanyu Xuexi (Chinese Language Learning)*.
- Cinque, Guglielmo. 1976. mica. *Annali della Facoltà di Lettere e Filosofia dell’Università di Padova*: 101–112.
- Cinque, Guglielmo. 1999. *Adverbs and Functional Heads: A Cross-Linguistic Perspective*. Oxford Studies in Comparative Syntax. New York: Oxford University Press.
- Ernst, Thomas. 1995. Negation in Mandarin Chinese. *Natural Language & Linguistic Theory* 13. Springer: 665–707.
- Faller, Martina. 2002. Semantics and Pragmatics of Evidentials in Cuzco Quechua. Ph.D. thesis, Stanford University.
- Frana, Ilaria, and Kyle Rawlins. 2019. Attitudes in discourse: Italian polar questions and the particle mica. *Semantics and Pragmatics* 12: 1–55.
- Grano, Thomas. 2015. *Control and Restructuring*. 1st ed. Oxford University Press/Oxford.
- Horn, Laurence R. 2001. *A Natural History of Negation*. The David Hume Series. Stanford, Calif: CSLI.
- Hu, Yong. 2008. Yuqi fuci “bing” de yufa gongneng yu fouding [The grammatical functions of the modal adverb “bing” and negation]. In *Yufa Yanjiu He Tansuo (Shisi) [Grammatical Research and Explorations (Fourteen)]*. Commercial Press.
- Huang, C.-T. James. 1988. Wǒ pǎo de kuài and Chinese phrase structure. *Language* 64: 274.

- Karttunen, Lauri. 1973. Presuppositions of compound sentences. *Linguistic Inquiry* 4. The MIT Press: 169–193.
- Kasimir, Elke. 2008. On “nicht...sondern...” (contrastive ‘not...but...’). *Working Papers of the SFB 632, Interdisciplinary Studies on Information Structure (ISIS)* 5: 107–151.
- Li, Jinrong. 2014. Qingli zhizhong yu yuliao zhiwai: Tan “bing” he “you” de yufa yiyi [Within and beyond the expectation: On the semantics of “bing” and ‘you’]. *Hanyu Xuexi (Chinese Language Learning)*: 12.
- Liu, Cheng. 2010. Shixi qianxiang yinhan de “you” zi ju [Attempt at analysis of the “you” sentence implied in the preceding clause]. *Huawen Jiaoxue yu Yanjiu (Journal of College of Chinese Language and Culture of Jinan University)*: 8.
- Ma, Zhen. 2001. Biao jiaqiang fouding yuqi de fuci “bing” and “you” -- jiantan ciyu shiyong de yuyi beiing [Adverbs “bing” and “you” that express intensive negative tone -- also on the semantic background of word use]. *Shijie Hanyu Jiaoxue (Chinese Teaching In The World)*: 7.
- Mari, Alda. 2024. Scope ambiguities in future questions: reflection and queclamative with Italian MICA. In *Proceedings of Sinn und Bedeutung* 28.
- Murray, Sarah E. 2017. *The Semantics of Evidentials*. Vol. 1. Oxford University Press.
- Nie, Xiaoli. 2021. “You” zi foudingju de fumian shili lichang biaoda gongneng [The negative factual stance expression function of “you” negative sentences]. *Yuyan Jiaoxue yu Yanjiu (Language Teaching and Linguistic Studies)*.
- Papafragou, Anna. 2006. Epistemic modality and truth conditions. *Lingua* 116: 1688–1702.
- Peng, Xiaochuan. 1999. Fuci “bing”, “you” yongyu fouding xingshi de yuyi, yuyong chayi [Differences in semantics and pragmatics between the adverbs “bing” and “you” when applied to the negative form]. *Journal of Central China Normal University (Humanities and Social Sciences)*: 57–64. CNKI.
- Potts, Christopher. 2004. *The Logic of Conventional Implicatures*. Oxford University Press.
- Potts, Christopher. 2015. Presupposition and implicature. In *The Handbook of Contemporary Semantic Theory*, ed. Shalom Lappin and Chris Fox, 1st ed., 168–202. Wiley.
- Repp, Sophie. 2013. Common ground management: Modal particles, illocutionary negation and verum. In *Beyond Expressives: Explorations in Use-Conditional Meaning*, ed. Daniel Gutzmann and Hans-Martin Gärtner, 231–274. BRILL.
- Roberts, Craige. 1996. Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5.
- Shi, Jinsheng. 2005. “You”, “ye” de bianbo yuqi yongfa jiqi yufahua [The refutative tone use of “you” and “ye” and its grammaticalization]. *Shijie Hanyu Jiaoxue (Chinese Teaching In The World)*: 54–62.
- Shi, Xiyao. 1990. Fuci “you” de yuyi jiqi wangluo xitong [The semantics and network system of the adverb ‘you’]. *Yuyan Jiaoxue yu Yanjiu (Language Teaching and Linguistic Studies)*: 11.
- Toosarvandani, Maziar. 2014. Contrast and the structure of discourse. *Semantics and Pragmatics* 7.
- Umbach, Carla. 2004. On the notion of contrast in information structure and discourse structure. *Journal of Semantics* 21: 155–175.
- Umbach, Carla. 2005. Contrast and information structure: A focus-based analysis of but. *Linguistics* 43.
- Ussery, Cherlon, Lydia Ding, and Yining Rebecca Liu. 2016. The typology of Mandarin infinitives. *Proceedings of the Linguistic Society of America* 1: 28.

- Wang, Minghua. 2001. Yong zai foudingci qianmian de “bing” yu zhuanzhe [‘Bing’ used before negation and counterexpectation]. *Shijie Hanyu Jiaoxue (Chinese Teaching In The World)*. 19–24.
- Wei, Wei. 2020. Discourse Particles in Mandarin Chinese. Ph.D. thesis, University of Southern California.
- Wen, Guifang, and Xiaojun Li. 2019. Goushi “youbu/mei Xp” de gongneng jiqi xingcheng [The function and formation of structural ‘youbu/mei Xp’]. *Yuyan Jiaoxue yu Yanjiu (Language Teaching and Linguistic Studies)*.
- Wen, Suolin. 2009. Yuqi fuci “bing” de yufa yiyi [The semantics of the mood adverb ‘bing’]. *Yuwen Yanjiu*.
- Winter, Yoad, and Mori Rimon. 1994. Contrast and implication in natural language. *Journal of Semantics* 11: 365–406.
- Wu, Zhongwei. 1999. Lun “youbu p, ~q” zhong “you” de yiyi [On the meaning of “you” in ‘youbu p, ~q’]. *Hanyu Xuexi [Chinese Language Learning]*.
- Yang, Bin. 2008. “Bing”, “you” yu foudingci lianyong de duo jiaodu fenxi [The multi-angle analysis of “bing” and “you” and their co-occurrence with negation]. *Dangdai Xiucixue [Contemporary Rhetoric]* 000: 61–65.
- Yin, Hongbo. 2011. “Bingbu” zhong “bing” de gongneng [The function of “bing” in ‘bingbu’]. *Beihua Daxue Xuebao (Shehui Kexue Ban) (Journal of Beihua University (Social Sciences))* 12.
- Yuan, Mengxi, and Mingming Liu. 2025. Cong duowei yuyi shijiao zai tan yuqi fuci “bing” yu “you” de yitong [Distinctions between discourse adverbs bing and you in multi-dimensional semantics]. *Yuyan Jiaoxue yu Yanjiu [Language Teaching and Linguistic Studies]* 233: 91–102.
- Yuan, Yulin. 2023. Duoshengxing biaoji “bing” de fanxiang binglie yiyi ji qi lishi lai yuan [A study on Chinese polyphonic marker bing: Its sense of reverse coordination and multiple historical origins]. *Hanyu Xuebao [Chinese Linguistics]*.
- Zhang, Jingyu, and Jianing Liu. 2010. Hanyu jianjie fouding jujue jushi “youbu/meiyou” de yuyi beijing he shiyong tiaojian [The semantic background and use conditions of the indirect negative denial structure “youbu/meiyou” in Chinese]. *Hanyu Xuexi (Chinese Language Learning)*.
- Zhang, Jingyu, and Junli Yan. 2015. Zhijie he jianjie fouding jujue: “Bing” he “you” de yupian xianjiexing [Direct and indirect negative denial: The discourse cohesion of “bing” and you’]. *Jilin Shifan Daxue Xuebao: Renwen Shehui Kexueban (Jilin Normal University Journal: Humanities & Social Science Edition)* 000: 21–26.
- Zhou, Xuan. 1999. Mandarin negations as negative aspectual elements. In *Proceedings of NELS*. Vol. 29.